Assignment 1

Name :- Sandeep Prasad Yadav

Reg no. :- 19251149

Course :- Theory of Computation.

Course Id :- CSA1311

Q1 Design a DFA to accept strings that start and end with the same symbols over $\Sigma = 1$, Also show that 0101001 is accepted.

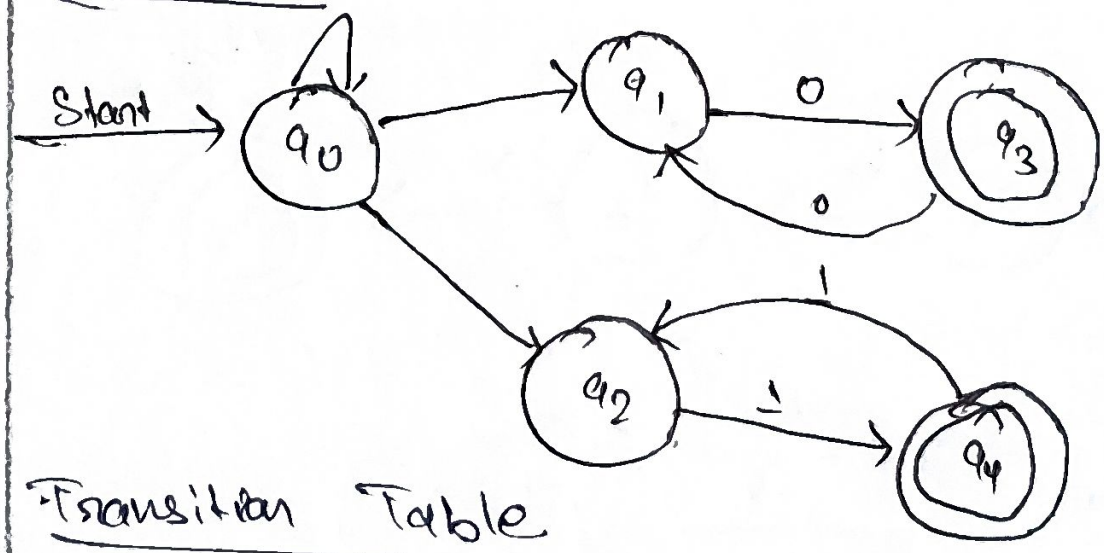
Soln

Formal Definition

$M = (Q, \Sigma, \delta, q_0, f)$

- $Q = (q_0, q_1, q_2, q_3, q_4)$
- $\Sigma = (0, 1)$
- Initial state : q_0
- Final state : $f = (q_1, q_2)$

Diagram:



Transition Table

state	0	1
→ q_0	q_1	q_3
* q_1	q_1	q_2
q_2	q_1	q_2
q_3	q_4	q_3
* q_4	q_2	q_4

Given String:

$w = 0101001$

$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \xrightarrow{0} q_1 \xrightarrow{0} q_1$

final state = q_1 Accepted

Q. Design an NFA to accept strings having 101 as a sub string.

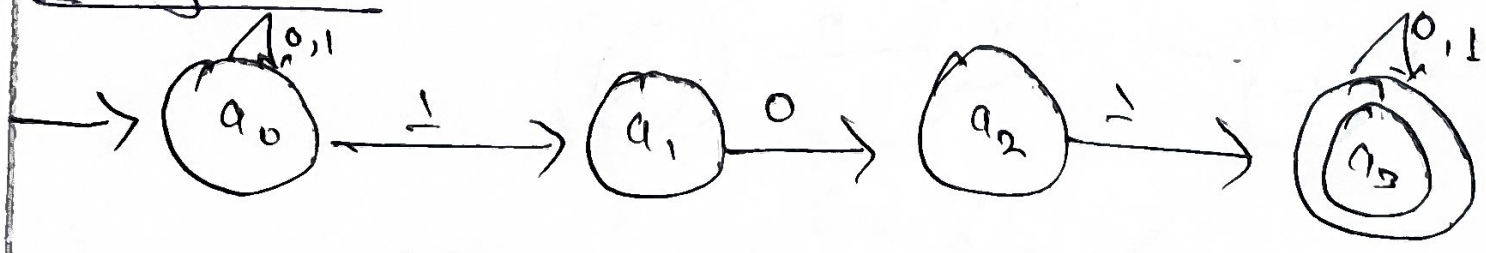
Soln.

Formal defination.

$$M = (Q, \Sigma, \delta, q_0, F)$$

- $Q = (q_0, q_1, q_2, q_3)$
- $\Sigma = (0, 1)$
- Initial state: q_0
- Final state: $F = (q_3)$

Diagram.



Transition table

State	0	1
→ q_0	(q_0)	(q_0, q_1)
q_1	(q_2)	$\emptyset$
q_2	$\emptyset$	(q_3)
* q_3	$(q_3$	(q_3)

given :- 1010 $q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \xrightarrow{0} q_3 \xrightarrow{0} q_3$

Q3 Prove using mathematical induction.

Given:

$$\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$

Base case (n=1)

L.H.S :

$$1(1+1) = 2$$

R.H.S :

$$\frac{1(2)(3)}{3} = 2$$

L.H.S = R.H.S

Hence, true for n=1.

Induction Hypothesis.

Assume the statement is true for n = k.

$$\sum_{i=1}^{k} i(i+1) = \frac{k(k+1)(k+2)}{3}$$

Induction Step

For n = k+1,

$$\sum_{i=1}^{k+1} i(i+1)$$

$$= \sum_{i=1}^{k} i(i+1) + (k+1)(k+2)$$

Using the induction hypothesis.

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

Taking common term

$$= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3}$$

$$= \frac{(k+1)(k+2)(k+3)}{3}$$

This is the required formula for $n = k+1$

Hence, $n = 1$ & $n = k+1$ true

$$\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$

Proved

4. Convert the given NFA into an equivalent DFA.

Given: $\Sigma = (a, b)$

The NFA accepts strings that start and end with the same symbol., with length at least 2.

NFA State:

- Initial State: q_0
- Final State: q_2
- Other State: q_1, q_3

NFA Transition Table

State	a	b
→ q_0	(q_1)	(q_3)
q_1	(q_1, q_2)	(q_1)
q_3	$q(\emptyset)$	(q_3, q_2)
* q_2	$\emptyset$	$\emptyset$

Subset Construction

let, $A=(q_1)$, $B=(q_1)$, $C=(q_3)$, $D=(q_1,q_2)$

$E=(q_3,q_2)$

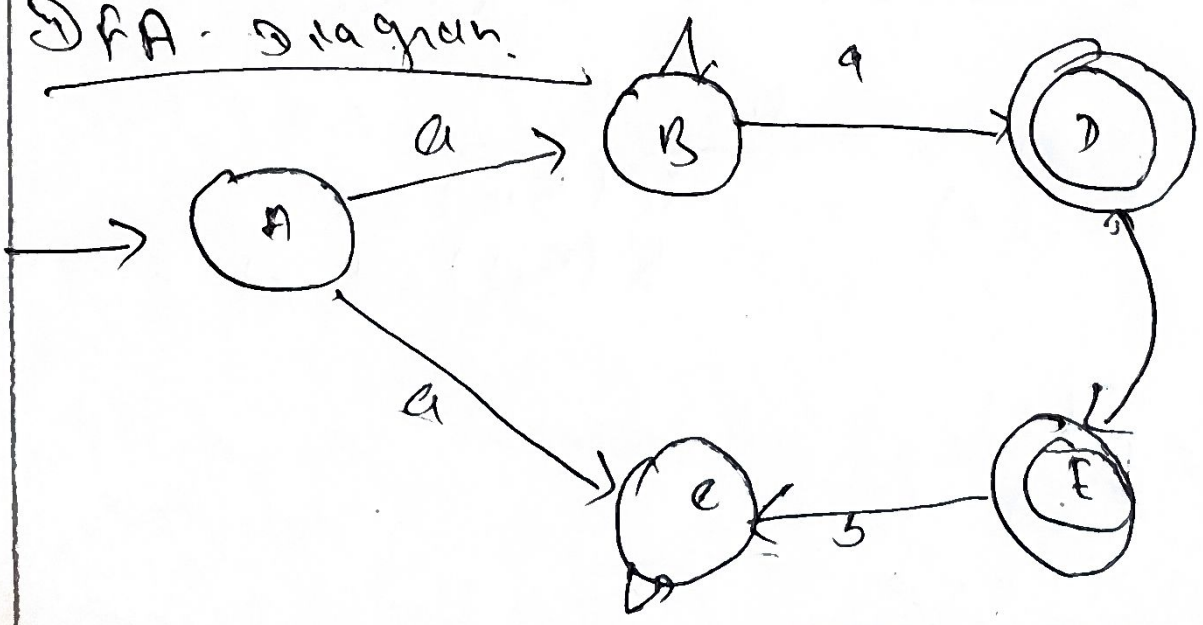
Equivalent DFA Transition Table.

DFA State	a	b.
→ A	B	C
B	D	B
C	C	E
*D	D	B
*E	C	F

initial State : A

final state : D, E

DFA - Diagram.



Q5 Find ε-closure & construct equivalent NFA without ε - transitions.

given NFA -

$Q = (A, B, C)$

Alphabet : $\Sigma = (0, 1)$

Initial State : C Final State : C.

Transitions :

- A has a loop on 0
- A → B on ε & 0
- A → C on 0.
- B has a loop on 1.
- B → C on ε.
- C has loops on 0 & 1.

Find ε - closer.

ε - closer(A) = (A, B, C)

ε - closer(B) = (B, C)

ε - closer(C) = (C)

Remove ε - transitions.

Using:

$$\delta'(q, x) = \varepsilon\text{-closure}(\delta(\varepsilon\text{-closure}(q), x))$$

for A,

$\delta'(A, 0) = (B, C)$

$\delta'(A, 1) = (A, B, C)$

for B

$\delta'(B, 0) = C$

$\delta'(B, 1) = B, C)$

for C

$\delta'(C, 0) = (C)$

$\delta'(C, 1) = (C)$

Table

State	0	1
→ * A	(B, C)	(A, B, C)
* B	(C)	(B, C)
* C	(C)	(C)

Initial State : A

Final State : A, B, C

L = (0, 1)*